Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/07/2025
PI Name:	Pathak, Siddhartha Ph.D.		
Project Title:	Structure, Properties, and Performance of Solder Joints in Terrestrial vs. Reduced-Gravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved in fall 2020 to Iowa State University from University of Nevada, Reno.		
Project Type:	Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2021 Physical Sciences NNH21ZDA014N-PSI: Use of the NASA Physical Sciences Informatics System – Appendix G
Start Date:	01/01/2023	**End Date:**	12/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Napolitano, Ralph Ph.D. (Iowa State University, Ames)		
Grant/Contract No.:	80NSSC23K0279		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The investigation proposed here combines experiments and modeling to elucidate the fundamental mechanisms, phenomenology, and process conditions that govern the integrity and performance of solder joints produced in terrestrial vs. reduced gravity environments, such as the microgravity conditions on board the International Space Station (ISS). The technical research program plans to utilize solder samples from the In-Space Soldering Investigation (ISSI) experiments from the Physical Sciences Informatics (PSI) repository, as well as expand into other non-ISSI solder compositions, and combine space- and ground-based experiments with advanced 3D materials characterization, micromechanical testing, and mesoscale modeling. In particular, the project addresses the formation and persistence of porosity through the reflow/filling/freezing processes and the deleterious effects on microstructure and mechanical properties of the solder joint. It has been established that porosity arising from flux volatilization, which is dispersed and expelled from the solder joint under terrestrial gravity, may become entrapped within the freezing solder material under microgravity conditions, given the absence of buoyancy-driven convection. Our overall goals are (i) to advance the current qualitative understanding of this phenomenon into the realm of alloy/process-specific quantitative description and prediction, and (ii) to examine the effects of mechanically and acoustically stimulated flow patterns while assessing their potential effectiveness as porosity mitigation strategies for solder-based fabrication processes in space. Considering a range of potential applications and materials, 3 solder alloys will be investigated, including the ISSI lead-based (Pb-Sn) solders, as well as lead-free (Sn-Ag-Cu and Sn-Au) solders, which have recently shown promise for high-performance joint applications due to their thermal/electrical conductivities and excellent corrosion/fatigue resistance.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:

The ongoing project is unique in that it combines both experimental and computational research on solder samples from the microgravity and ground based ISSI experiments, along with a comparative study on fresh solder samples. This project has the following research impact: (i) Our project objectives are a timely alignment with current NASA initiatives related to metals joining in reduced gravity environments. The three solder alloys for this work were chosen after extensive discussions with our NASA collaborators, and these alloy systems are currently being considered at NASA as possible candidates for use in the NASA Mars Rover sample return missions. (ii) We also anticipate that the results from this project will inform experiment design responsive to the (future) NASA proposal calls such as the NASA Artemis program, which seeks to land the first woman and next man on the Moon. (iii) The insights gained from this work can be extended to study effects of fluid interfaces with surface tension variations that are also common for many spaceflight-enabling technologies (e.g., boiling, heat transfer, combustion, welding, brazing, or soldering). We envision that this project will contribute towards advanced understanding of multiple processes that typically rely on gravity-driven separations in the terrestrial environment, such as the removal of gas bubbles from a liquid, which must be accomplished by other means in microgravity environments. (iv) the developed gravity-level-specific protocols (GLSPs) for several configurations can be used by future scientists and flight personnel for studying metal joining techniques such as welding, brazing, or soldering under reduced gravity.
Additionally, this project studies the reliability of the solder joints under the extremely elevated and cryo temperatures experienced during the moon and Mars rover missions, which will have a high impact on the success of future deep space exploration. NASA is currently planning future landings to the Moon's south pole, which is filled with deep, dark craters and has been sunless for billions of years. In these areas, the temperature can drop to -203 degrees C, which makes remote observation from Earth difficult and presents problems for operating sensitive equipment in these regions. For example, India's Chandrayaan-3 mission landed Vikram lander and Pragyan rover near the moon's southern pole to study the mineral composition of the lunar surface and collect data to look for signs of frozen water (essential for rocket fuel). After spending a little under two weeks exploring the promising area, the rover had been "set into sleep mode" with its scientific instruments turned off. However, according to ISRO, it could not survive the harsh lunar night, and its hardware probably succumbed to the moon's freezing conditions. More recently, JAXA's Smart Lander for Investigating Moon (SLIM) landed near the Shioli crater, south of the lunar equator, and as of today SLIM has entered a two-week dormancy period during the long lunar night. According to JAXA, SLIM is not designed for the harsh lunar night. For the SLIM probe to wake up again during the next lunar day, however, its electronic components must survive nighttime temperatures of around -130 degrees C. Reliability of electronic components and metal joints under these harsh extremes of temperature in deep space is an urgent issue that needs more investigation. Therefore, the insight gained from this research will ensure the reliability of the electronic components used in rover missions by studying the properties of different solder compositions under extremely elevated and cryo temperatures experienced during space missions and assist in the development of new solder alloys that can withstand harsh temperatures. Furthermore, this research will directly support NASA's initiative of using a distributed, non-heated architecture for the rover and spacecraft electronics that does not require heater power or a "warm electronics box" (WEB).

Project Overview
This research provides quantitative correlations between solidification and flow characteristics, processing conditions, microstructure, and porosity for solder joints produced in terrestrial vs. microgravity conditions (onboard the International Space Station (ISS)). Furthermore, this project focuses on analyzing solder joint properties under extreme conditions of elevated and cryogenic temperatures in space. Soldering is the predominant method used to create electrical/mechanical joints and mechanical/pressure joints and is typically a rapid process in which the solder material is molten for only a few seconds. Solder joint porosity is a common but undesirable feature naturally arising from the use of fluxes and is more insidious in soldering joints formed under microgravity conditions. Under the presence of gravity, voids and bubbles are removed from the solder joints due to presence of buoyancy force and vigorous liquid solder mixing. In the absence of gravity, fluid motion is greatly reduced, driven mainly by solidification volume change, thermally induced density gradients, and Marangoni effects. With slower mixing, voids and bubbles are not swept away and become entrapped in the interior of the solder joint upon solidification. This entrapped porosity dramatically degrades the thermal and electrical properties of the solder and severely reduces the mechanical integrity of the joint. Therefore, as part of the ongoing endeavors to establish an electronics and mechanical joint repair capability for extended space missions, this project seeks to advance the current understanding of fundamental mechanisms, phenomenology, and process conditions that govern the integrity and performance of solder joints produced in terrestrial vs. reduced gravity environments. Additionally, solder joints in the space environment are subjected to large temperature variations such as – 120 ºC to +25 ºC for the Mars missions, and – 157 ºC to +121 ºC outside ISS. However, as deep space exploration missions become longer and more distant, the solder and other metal joints' current reliability is inadequate for the harsh conditions they encounter.

This project focused on the following objectives to address the reliability of solder joints in space: 1) Characterizing the effects of the lack of Earth's natural convective flow and buoyancy on solder-joint formation onboard the ISS using the 40Pb-60Sn solders from the In-Space Soldering Investigation (ISSI), 2) Investigating the effects of elevated and cryogenic temperature conditions in space on solder properties. Considering a range of potential applications and materials, three solder alloys were investigated, including the ISSI lead-based (40Pb-60Sn) solders, off-eutectic 50Pb-50Sn solders, and lead-free (95.5Sn-3.8Ag-0.7Cu) solders, which had recently shown promise for high-performance joint applications due to their good thermal and electrical conductivities, as well as excellent corrosion and fatigue resistance.

Our team conducted the microstructural characterization of four microgravity-processed solders from the ISSI study, one ISSI terrestrial solder, and a third set of freshly made terrestrial 40wt%Pb-60wt%Sn solder to demonstrate the effects of the absence of buoyancy and natural convection on void formation in microgravity. The effect of void-induced Marangoni convection on the solder microstructure was demonstrated using the microgravity wire-feed solder. Additionally, the micromechanical response of solders processed in terrestrial and microgravity environments was investigated under extreme elevated and cryogenic temperature conditions (ranging from –157ºC to +121ºC, as experienced outside the ISS). Under cryogenic conditions, the study investigated the impact of the β-to-α phase transformation in Sn below 13ºC, including associated volume changes, internal stress development, and the ductile-to-brittle transition observed in various Sn-based solder alloys. At room temperature, the effects of accelerated aging on the solders' microstructure and mechanical properties were explored.

Major Goals/Objectives

The major focus of this project was to elucidate the fundamental mechanisms, phenomenology, and process conditions that govern the integrity and performance of solder joints produced in terrestrial vs. reduced gravity environments, such as the microgravity conditions onboard the International Space Station (ISS). The primary objectives included advancing the current qualitative understanding of porosity formation and persistence in solder joints into the realm of alloy/process-specific quantitative description and prediction and evaluating the influence of elevated and cryogenic temperatures on the microstructural and mechanical properties of solder joints, with emphasis on Sn-based alloys.

Leveraging solder samples from the In-Space Soldering Investigation (ISSI) and other non-ISSI compositions, the project employed a combination of advanced experimental techniques and computational modeling. These approaches were designed to characterize the impacts of void-induced microstructural variations, extreme thermal cycling, and phase transformations, addressing critical reliability challenges for soldering in space environments. The project aimed to address critical scientific gaps related to solder joint reliability in the unique microgravity and thermal conditions of space, with the ultimate goal of providing quantitative insights for designing more robust soldering techniques. While this proposal focused on understanding the fundamental phenomena, it also laid a solid scientific foundation for future advancements in space fabrication and repair technologies. This work contributes directly to NASA's goals of improving in-space manufacturing and repair capabilities while establishing a strong foundation for future innovations in soldering processes tailored to the harsh conditions of deep-space exploration.

Accomplishments under Goals

The research program achieved several significant accomplishments in advancing the understanding of solder behavior in terrestrial vs.

Task Progress:

microgravity environments, with a focus on enhancing their reliability for space applications. This study analyzed near eutectic 40Pb-60Sn solder samples from the In-Space Soldering Investigation (ISSI), conducted in microgravity aboard the International Space Station (ISS) and under terrestrial conditions at NASA Marshall. The ISS experiments were performed by taking pieces of silver-coated copper wire and bending and twisting them into multiple test coupon configurations. The selected lengths of 40wt%Pb-60wt%Sn wire were cut and 1) wrapped around the test coupons and melted using the soldering iron – Wire wrap solder and 2) fed on the similar test coupons that were heated by the soldering iron – Wire feed solder (Fig.1). The ground experiments involved conductively melting 10 cm lengths of 40Pb-60Sn (3.3 wt% rosin flux) solder onto a loop turned into the silver coated copper wire. For more details about the ISSI experiments, refer to: Background: In-Space Soldering Investigation (ISSI 2003-04) section in this report. Additionally, near-eutectic 40Pb-60Sn, off-eutectic 50Pb-50Sn, and 95.5Sn-3.8Ag-0.7Cu terrestrial solder samples (named ISU terrestrial) were prepared and examined in Dr. Pathak's research lab at Iowa State University, Ames. For a detailed list of samples and tasks, refer to the Outline of Executive Summary of this report. Key accomplishments include:

Accomplishments from the study on near-eutectic 40Pb-60Sn solder (microgravity and terrestrial):

Samples analyzed: ISSI microgravity 40Pb-60Sn wire warp solder (Microgravity test 2 sample 4, 5 and 10) ISSI microgravity 40Pb-60Sn wire feed solder (Microgravity test 4 sample 16 – MT4S16) ISSI terrestrial 40Pb-60Sn wire feed solder ISU Terrestrial 40Pb-60Sn wire wrap and wire feed solder (prepared in Dr. Pathak's lab) ISU Terrestrial 40Pb-60Sn Reflow Spreading solder (prepared in Dr. Pathak's lab) • Demonstrated the effect of natural convection and buoyancy on void size in 40Pb-60Sn solder formed under microgravity vs. terrestrial conditions using non-destructive 3-D tomography. Results demonstrated that the occurrence of larger voids (as large as 1 mm in diameter) only happens in microgravity conditions. • Developed a quantitative framework for describing and predicting porosity formation and persistence in 40Pb-60Sn solder under microgravity vs. terrestrial conditions, addressing the role of reduced buoyancy-driven convection and flux volatilization. ISSI microgravity wire wrap 40Pb-60Sn solder had ~13× more voids than ISSI terrestrial 40Pb-60Sn solder, while ISSI microgravity wire feed 40Pb-60Sn solder had 4× more voids than ISSI wire wrap 40Pb-60Sn solder. • Studied the effect of Marangoni convection in microgravity 40Pb-60Sn solder under the presence of a large free surface (large void, greater than 1 mm) on solder microstructure (primary dendrite and small void accumulation). The average Pb-particle size near the hot site was 6.1 ± 5 µm and near the cold site (expected to be at a lower temperature than the hot site) was 4.8 ± 5 µm which was caused by the differential cooling rate in the colder and hot sites and accumulation of Pb-rich dendrite in the hot region. • Quantified the effect of microstructure, influenced by void-induced Marangoni convection, on microgravity solder properties. We demonstrated that Inhomogeneity (accumulation of Pb-rich dendrites at the solder joint) resulting from Marangoni flow caused lower strength (hardness ~ 0.11 GPa) of the solder at the hot site than at the colder site (hardness ~ 0.15 GPa). • The aging study on terrestrial 40Pb-60Sn solder showed a hardness decrease from 0.19 GPa on day 1 to 0.16 GPa on day 483, accompanied by Pb particle growth from 1.28 µm to 2.45 µm over the same period. • Investigated the β-Sn to α-Sn phase transformation below 13°C on pure Sn single crystal and terrestrial 40Pb-60Sn solder using high-throughput spherical nanoindentation, elucidating the effect of associated volume changes and internal stresses on mechanical performance in cryogenic environments. • Investigated the effects of ductile to brittle transition on terrestrial 40Pb-60Sn solder properties via in-situ SEM micropillar compressions. Plastic strain at instability under compression in 40Pb-60Sn decreased from $\varepsilon = 0.109$ at RT to $\varepsilon = 0.06$ below -85°C, suggesting a ductile-to-brittle transition near this temperature.

Accomplishments from the study of terrestrial off-eutectic 50Pb-50Sn solder:

Sample analyzed: ISU Terrestrial 50Pb-50Sn solder sample 1 to 3 (prepared in Dr. Pathak's lab) • Aging study on terrestrial 50Pb-50Sn demonstrated that the hardness dropped from 0.18 to 0.14 GPa, and Pb particle size grew from 0.45 to 0.8 µm over 89 days. Accomplishments from the study on terrestrial 95.5Sn-3.8Ag-0.7Cu solder: Sample analyzed: ISU Terrestrial 95.5Sn-3.8Ag-0.7Cu solder sample 1 (prepared in Dr. Pathak's lab) • In-Situ SEM micro-pillar compression on ISU terrestrial 95.5Sn-3.8Ag-0.7Cu solder demonstrated that plastic strain at instability under compression dropped from $\varepsilon = 0.06$ at RT to $\varepsilon = 0.03$ at -85°C and lower temperatures, signaling a ductile-to-brittle transition around this temperature. • The deformation behavior was correlated with microstructure across all temperatures. Group 3 (eutectic mixture of Ag3Sn + Sn) demonstrated a sharp decrease in the plastic strain at instability as the temperature drops below room temperature (RT), probably caused by the high fraction of the Ag_3Sn intermetallic phase that significantly influences the mechanical response. At cryogenic temperatures, Ag_3Sn is much stiffer and more brittle than the Sn-rich phase. As temperature decreases, deformation becomes increasingly localized at phase boundaries, reducing plasticity and accelerating the ductile-to-brittle transition. • In contrast, Group 2a (eutectic Ag3Sn + Sn + Sn-rich dendrite) showed a gradual decrease in plastic strain with decreasing temperature, as its balanced mix of Sn-rich and eutectic phases allowed for some residual plasticity, unlike the sudden drop observed in Group 3. Group 2b (Cu5Sn6 + group 2a) exhibited the lowest plastic strain across all temperatures due to the presence of Cu_6Sn_5 intermetallic compounds, which increased stiffness and restricted plastic deformation, leading to early instability.

Objectives and Outcomes Year 1: Sample analyzed: • ISSI M 40Pb-60Sn Test 2 Sample 4 • ISSI M 40Pb-60Sn Test 4 Sample 16 • ISSI 40Pb-60Sn Terrestrial Sample 2 • ISSI 40Pb-60Sn Terrestrial Sample 3 • ISU 40Pb-60Sn Terrestrial Sample 1

The first year of this project was focused on studying the near-eutectic 40Pb-60Sn (terrestrial and microgravity) and off-eutectic 50Pb-50Sn (terrestrial) solder compositions. This study focused on understanding the role of reduced buoyancy forces and natural convection in void formation and microstructural inhomogeneity in 40Pb-60Sn solder under microgravity conditions, alongside the effects of room temperature aging on ISSI microgravity 40Pb-60Sn, terrestrial 40Pb-60Sn and terrestrial 50Pb-50Sn solder properties. Additionally, the study aimed to quantify the influence of Marangoni flow and thermal convection on void dynamics and phase segregation in ISSI microgravity 40Pb-60Sn solder. The study demonstrated that microgravity conditions significantly increased void fraction (ISSI microgravity wire wrap solder had ~13× more voids than terrestrial solder) and size in solder joints due to the absence of buoyancy-driven convection, with Marangoni flow contributing to void and Pb-rich dendrite accumulation in high-temperature regions. Mechanical testing revealed ductile-to-brittle transitions in ISU terrestrial 40Pb-60 Sn solder under cryogenic conditions at temperatures at and below -85 oC, while aging studies highlighted grain coarsening (Pb particle size =1.28 µm on day 1 to 2.45 µm on day 483) and hardness reduction (0.19 GPa on day 1 to 0.15 GPa on day 483) in 40Pb-60Sn solder alloys. Please see the year one report for further details of this section.

Year 2:

Sample analyzed: ISSI M 40Pb-60Sn Test 2 Sample 10 ISSI M 40Pb-60Sn Test 4 Sample 16 ISU 40Pb-60Sn Terrestrial Sample 3 ISU 50Pb-50Sn Terrestrial Samples 1, 2, and 3 ISU 95.5Sn-3.8Ag-0.7Cu Terrestrial Sample 1

The second year of this project focused on 1) quantifying the effects of void-induced Marangoni convection and thermal gradients on ISSI microgravity 40Pb-60Sn solder strength and 2) evaluating the influence of elevated and cryogenic temperatures on Sn-based terrestrial solders (40Pb-60Sn, 50Pb-50Sn and 95.5Sn-3.8Ag-0.7Cu) properties, including ductile-to-brittle transition and phase transformations.

The study revealed that void-induced Marangoni convection in microgravity significantly reduces solder hardness near high-temperature regions due to the localized accumulation of softer Pb-rich phases, highlighting the critical need to develop soldering processes that mitigate microstructural inconsistencies to ensure the structural integrity of solder joints in space environments. Cryogenic temperature studies on ISU terrestrial 40Pb-60Sn solder using high throughput spherical nanoindentation stress-strain analysis revealed that 40Pb-60Sn solder exhibits increased yield strength from 0.18 GPa (0.05 offset yield strength) at RT to 0.32 GPa at -120°C, while pure Sn shows unusual softening at -120°C, likely due to the β-to-α phase transformation, emphasizing the need for alloy-specific reliability assessments under extreme conditions. The micro-pillar compression experiments revealed a significant ductile-to-brittle transition in ISU terrestrial 40Pb-60Sn solder below -85°C, characterized by decreased plastic strain at instability from 0.109 at RT to 0.062 at -85°C and deformation concentrated along Pb-rich and Sn-rich phase boundaries. Micropillar compression experiments on ISU terrestrial 95.5Sn-3.8Ag-0.7Cu solder at cryogenic temperatures revealed a significant increase in stress at instability which changed from 78 MPa at RT to 153 MPa at -145°C, with a noticeable saturation in instability stress below -85°C and a reduction in the plastic strain at instability (from 0.06 at RT to 0.03 at -85°C), primarily attributed to a ductile-to-brittle transition. Deformation mechanisms in 95.5Sn-3.8Ag-0.7Cu varied with temperature and percentage of various phases in the different pillars, with deformation predominantly progressing along Sn-rich dendrites at -40°C, shifting to eutectic and intermetallic interfaces below -85°C, highlighting the critical role of phase boundaries in cryogenic

mechanical behavior.

Dissemination

Summary of Technical Dissemination: Research undertaken and supported by this grant was completed at Iowa State University Ames by graduate and undergraduate researchers. Scientific products were finalized and disseminated through professional society technical presentations, invited academic, and technical seminars, and student posters at several academic and society events. While most technical presentations were given by the PIs, graduate researchers supported by this project had many opportunities for presentation and publishing. At the time of this final report's submission, four peer-reviewed publications are in the final stages of preparation for submission. Further, outcomes from research activities in this program were included as 1 invited talk and 12 conference talks, including graduate student Manish Kumar's one PhD dissertation and one more invited talk to be given after the completion of this grant.

Products:

2025 • Kumar M, Napolitano R, Pathak S, "Reliability of Terrestrial vs. Microgravity Solders under Extreme Elevated and Cryo Temperatures for Deep Space Exploration" , The Minerals, Metals and Materials Society (TMS), , Las Vagas, NV March 23–27, 2025. 2024 • Kumar M, Napolitano R, Pathak S, "Reliability of Terrestrial vs. Microgravity Solders under Extreme Elevated and Cryo Temperatures for Deep Space Exploration" Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), San Juan, Puerto Rico December 3-7, 2024. • Kumar M, Napolitano R, Pathak S, "Reliability of Terrestrial vs. Microgravity Solders under Extreme Elevated and Cryo Temperatures for Deep Space Exploration" International Space Station Research & Development Conference (ISSRDC), Boston, Massachusetts, July 29-August 1, 2024. • Invited: Kumar M, Napolitano R, Pathak S, "Structure and Properties of Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," Virtual NASA PSI User Meeting, March 2024. • Kumar M, Napolitano R, Pathak S, "Structure and Properties of Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," The Minerals, Metals and Materials Society (TMS), Orlando, Florida, March 3-7, 2024. • Kumar M, Napolitano R, Pathak S, "Structure and Properties of Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," Center for Advanced Non-Ferrous Structural Alloys IRB Review Meeting, Iowa State University, Ames, Iowa, October 22-23, 2024.

2023 • Kumar M, Napolitano R, Pathak S, "Structure and Properties of Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Washington, D.C, November 14-18, 2023. • Hellyer S, Ogden C, Kumar M, Napolitano R, Pathak S, "Studying the Effects of Aging on the Structure and Properties of Off-Eutectic 50wt%Pb-50wt%Sn Solder Joints for In-Space Applications," Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Washington, D.C, November 14-18, 2023. • Kumar M, Napolitano R, Pathak S, "Structure and Properties of the Terrestrial vs. Microgravity Solders under Extreme Conditions of Elevated and Cryo Temperatures," International Space Station Research & Development Conference (ISSRDC), Seattle, Washington, July 31-August 3, 2023. • Hellyer S, Ogden C, Kumar M, Napolitano R, Pathak S, "Studying the Effects of Aging on the Structure and Properties of Off-Eutectic 50wt%Pb-50wt%Sn Solder Joints for In-Space Applications," National Conference on Undergraduate Research (NCUR), Eau Claire, Wisconsin, April 13-15, 2023. • Kumar M, Jacob K, Napolitano R, Pathak S, "Structure and Properties of Pb-Sn Solders Produced in Terrestrial vs. Microgravity Environments," Center for Advanced Non-Ferrous Structural Alloys IRB Review Meeting, Colorado School of Mines, Golden, Colorado, September 27-28, 2023. • Kumar M, Napolitano R, Pathak S, "Structure and Properties of Solder Joints Produced in Terrestrial and Microgravity Conditions," The Minerals, Metals and Materials Society (TMS), San Diego, California March 19-23, 2023. • Kamp J, Madison N, Kumar M, Pathak S, "Structure and Properties of Pb-Sn and Pb-Free 95.5Sn-3.8Ag-0.7Cu Solders Produced in Terrestrial Environment," ISU Symposium on Undergraduate Research and Creative Expression, Ames, Iowa, April 20, 2023.

Scientific Manuscripts under preparation • Kumar, M., Azizi, G., Napolitano, R., Zaeem, M., Pathak, S., "Effect of Aging on Structure and Properties of the 40Pb-60Sn Solders for In-Space Applications." • Kumar, M., Napolitano, R., Pathak, S., "Structure and Properties of Solder Joints Produced in Terrestrial vs. Microgravity Conditions." • Kumar, M., Hellyer, S., Ogden, C., Kamp, J., Napolitano, M., Pathak, S., "Effect of Aging on Structure and Properties of the 50Pb-50Sn Solders for In-Space Applications." • Kumar, M., Napolitano, R., Pathak, S., "Solder Joint Reliability under Extremes of Elevated and Cryogenic Temperatures Experienced during Deep Space Exploration."

Honors and Awards

2024 • December 2024: Graduate student Manish Kumar's research was prominently featured in NASA's 2025 Science Calendar for the month of November. Hear Dr. Fox's thoughts on our research during the ASGSR conference: https:// Link to download the high-resolution NASA Science calendar: https:// • March 2024: Undergraduate researcher Soren Hellyer was awarded the TMS Acta Materialia Scholarship-2024 • February 2024: Manish Kumar was awarded the Brown Graduate Fellowship Award ($10,000) – Recognized by the Office of the Vice President for Research at Iowa State University for exceptional achievements and contributions to space science. • January 2024: TMS student travel award ($300) - to help defray the costs of attendance at the Mineral, Metals and Materials Society annual meeting (TMS-2024).

2023 • November 2023: Manish Kumar was awarded ASGSR Student Travel Award ($500) to help defray the costs of attendance at the American Society for Gravitational and Space Research (ASGSR-2023) conference. • August 2023: Manish Kumar was awarded Wayne G. Basler Scholarship ($5000) for 2023 in Materials Science and Engineering at Iowa State University • June 2023: Sigma Xi Grants ($1000) in Aid of Research 2023 - Sigma Xi's Committee on Grants-in-Aid of Research has approved Manish's research grant proposal titled" Solder Joints Under Extreme Conditions of Temperature and High Strain-rate for In-Space Applications," submitted for the March 15, 2023 application cycle. Among the 97 recipients of the Sigma Xi award this year, only seven were granted in the field of Engineering, and Manish was one of those seven recipients. You can read the press release about the recipients: https:// • April 2023: Noah Madison was awarded the Outstanding First-Year Honors Mentor Program (FHMP) Scholar Award – 2023 • March 2023: Joshua Kamp and Noah Madison were awarded First-Year Honors Mentor Program Grant-2023 • February 2023: Dr. Pathak was awarded the 2023 Iowa NASA EPSCoR Partnership Development Travel Grant ($7500) to initiate a new research collaboration on soldering under extreme temperature conditions with NASA International Space Station (ISS) implementation partners and scientists at NASA Marshall • January 2023: Soren Hellyer and Caleb M Ogden received the NASA – Iowa Space Grant Consortium Award ($5000). This award was intended to provide experiences to undergraduates that promote skills for STEM careers through hands-on activities (Project mentors: Manish Kumar, Dr. Sid Pathak)

Training Opportunity

Graduate researchers who were involved in this project (PhD student at ISU - Mr. Manish Kumar) got the opportunity to do the following training and experiences: • Received training at the Sensitive Instrument Facility (SIF) at Ames Laboratory, including SEM/FIB, EBSD, and in-situ SEM Hysitron PI-85 Picoindenter to perform both in-situ nanoindentation and micropillar compression. • Visited Bruker Inc., Eden Prairie, MN, to perform In-Situ SEM Nanoindentation experiments at cryo temperature and received training on TI-980 Triboindenter. • Received training on Micro-EDM machine to fabricate the micro-pillars and micro-tensile geometries. • Traveled to the Advanced Photon Source in Argonne National Laboratory for Non-destructive 3-D tomography experiments. • Traveled to the University of Wisconsin, Madison to fabricate the tensile gripper using the Plasma FIB.

Undergraduate students (Soren Hellyer, Caleb Ogden, Noah Madison, Joshua Kamp) were trained on the automatic mechanical polisher (Tegramin-25), microstructural characterization tool (Optical microscope and SEM), and nano-mechanical testing tool (TI-950 nanoindenter).

Bibliography Type:	Description: (Last Updated: 04/22/2025)
Conference Materials (Downloadable)	Kamp J, Madison N, Kumar M, Pathak S. "Structure and properties of Pb-Sn and Pb-Free Sn-Ag-Cu solders produced in terrestrial environment." SU Symposium on Undergraduate Research and Creative Expression, Ames, Iowa, April 20, 2023. , Apr-2023 NCUR_2023_Hellyer_Ogden_23_3_23_Final.pdf
Conference Materials (Downloadable)	Kumar M, Napolitano R, Pathak S. "Structure and properties of solder joints produced in terrestrial and microgravity conditions." The Minerals, Metals and Materials Society (TMS), San Diego, California, March 19-23, 2023. , Mar-2023 2023 ISSRDC_Poster_Manish_Kumar 2023-07-24.pdf
Conference Materials (Downloadable)	Kumar M, Jacob K, Napolitano R, Pathak S. "Structure and properties of Pb-Sn solders produced in terrestrial vs. microgravity environments." Center for Advanced Non-Ferrous Structural Alloys IRB Review Meeting, Colorado School of Mines, Golden, Colorado, September 27-28, 2023. , Sep-2023 2023 CANFSA_Poster_Manish Kumar 2023-03-28.pdf
Conference Materials (Downloadable)	Hellyer S, Ogden C, Kumar M, Napolitano R, Pathak S. "Studying the effects of aging on the structure and properties of off-eutectic 50wt%Pb-50wt%Sn solder joints for in-space applications." National Conference on Undergraduate Research (NCUR), Eau Claire, Wisconsin, April 13-15, 2023. , Apr-2023 NCUR_2023_Hellyer_Ogden_23_3_23_Final.pdf
Conference Materials (Downloadable)	Kumar M, Napolitano R, Pathak S. "Structure and properties of the terrestrial vs. microgravity solders under extreme conditions of elevated and cryo temperatures." International Space Station Research & Development Conference (ISSRDC), Seattle, Washington, July 31-August 3, 2023. , Jul-2023 2023 ISSRDC_Poster_Manish Kumar 2023-07-24.pdf
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